RP500 — firmware reflash

Technician brief · DigiTech RP500 · board **80-0145-A** · 2026-11-12

Read before you start

- **Voltage: 3.3 V. Never 5 V.**
- **Do not** touch **U19 (TAS1020B)** or its I²C EEPROM — that is the USB controller and is not part of this repair.
- **Do not** power the unit with **U26** removed.
- The only chip to reflash is **U26**. Nothing else on the board needs to change.

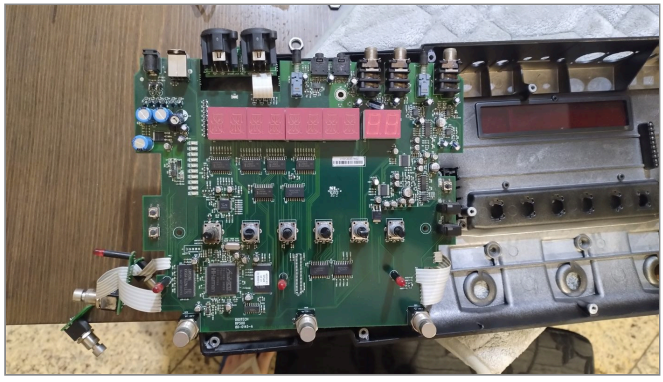
1. The chip

Item	Value
Board	DigiTech RP500, **80-0145-A**, component side
Reference	**U26**
Part number (under the white label)	**`SST29AL016J70TFI02`** (Spansion / SST)
Package	**TSOP-48, 0.5 mm pitch**
Size	**16 Mbit = 2 MB** (byte mode, 2M × 8)
Supply	**3.0–3.6 V (3.3 V)** — **never 5 V**
Speed	70 ns
Factory label	`72-0089-12  V1.20  RP500  CS.8CD9`

The chip sits between the **AudioDNA2** (large QFP, U25), the **Samsung K4S641632N** SDRAM (U24) and the row of **74HCT574** latches.



U26 — SST29AL016J flash (TSOP-48). The pin-1 dot is the reference.



Board 80-0145-A overview: U26 sits between the AudioDNA2 (U25), the SDRAM (U24) and the latches.

2. What the technician needs

Item	Requirement
Programmer	**XGecu T48** (recommended) or **TL866II Plus** with `SST29AL016J` / `SST29AL016` in its device list

Adapter	**TSOP48 to DIP48, 0.5 mm pitch** (not TSOP56)
Voltage	**3.3 V** (set the programmer/adapter to 3.3 V, not 5 V)
Tools	Hot-air station, flux, solder paste or fine iron, Kapton tape, magnifier
Software	The programmer's own software; no special tool needed

If the programmer's list has no `SST29AL016J`, use the **`SST29AL016`** entry (same JEDEC family) or a programmer that lists the Spansion `S29AL016J`.

3. The image files

The two `.bin` files must be supplied **alongside this PDF** (source: `09_builds/recovery_stock_1mb/`):

File	Size	SHA-256
`stock_1mb.bin`	1,048,576	`718fda7de9206838d5e4945d7cff0c890c3573dc68c81142df23f7dab765fa11`
`stock_2mb_padded_ff.bin`	2,097,152	`01978cbfbe20fc565a3c8100b415798855fd90e6bb6580ad193f5df5ac507d8d`

- `stock_1mb.bin` is the RP500 firmware, raw, **1 MB**.
- `stock_2mb_padded_ff.bin` is the same 1 MB followed by **1 MB of `0xFF`**, for programmers that insist on the chip's full 2 MB size.
- **Do not** use any `.DigBin` file directly; the programmer needs the raw `.bin` above.

First 16 bytes of the firmware (for checking):

```
EA 00 02 00 6B 53 91 DF 32 00 FF 02 23 4E 01 1F
```

Windows hash check:

```
certutil -hashfile stock_1mb.bin SHA256
```

4. Procedure

1. **Photograph U26 and note pin 1** before removal. Pin 1 is the small circular dot on the package; the board has a white corner marker.
2. **Desolder U26** with hot air. Protect the neighbouring SDRAM, DSP and latches with Kapton tape. Do not lift pads.
3. Clean the pads and place U26 in the **TSOP48 0.5 mm adapter**.
4. **Read the original chip completely (2 MB)** and save it as a backup (`orig_2mb.bin`). Check the first 16 bytes against the list above; the original may be the stock 2.0 image or the failed 2.2 image.
5. **Erase** the chip.
6. **Program** one of:
   - `stock_1mb.bin` at address **`0x000000`**, or
   - `stock_2mb_padded_ff.bin` at address **`0x000000`** if the programmer requires a full 2 MB file.
7. **Verify** the whole programmed area.
8. **Reinstall U26** in the original orientation (pin-1 dot aligned with the board marker).

9. Power on. The unit should boot the stock firmware and the display should show the normal RP500 start-up; USB should enumerate as **`VID_1210 / PID_0016`** (DigiTech RP500).

5. About the upper 1 MB

The RP500 updater addresses only **16 × 64 KB = 1 MB**, so the firmware lives in the **first half** of the 2 MB chip. The upper 1 MB is outside the updater's map.

- If the original upper 1 MB reads **all `0xFF`**, the padded file is correct.
- If the original upper 1 MB contains data, **do not overwrite it**: program the stock 1 MB into the lower half and restore the original upper half (merge `stock_1mb.bin + orig_2mb.bin[0x100000:0x200000]`).

6. Do not

- **Do not** use a 5 V setting.
- **Do not** touch **U19 (TAS1020B)** or its small I^2C EEPROM. That is the USB controller and is not part of this repair.
- **Do not** program the failed 2.2 S3b image (`be9676...`).
- **Do not** power the unit with U26 removed.

7. What "success" looks like

- Display: normal RP500 boot, no immediate power-down.
- USB: **`VID_1210 / PID_0016`**.
- The unit accepts the stock 2.0 firmware and behaves normally.

If it does not boot, re-check pin 1, solder joints, and the verify result before changing anything else.

8. Realistic cost

The only special item is the **TSOP48 0.5 mm adapter**. Any shop that does BIOS/EC/embedded-firmware work already has a T48 or TL866 and the adapter; the job itself takes 15–30 minutes. If they do not have the adapter, that is the only purchase — and it is still far cheaper than a new board or a pre-programmed chip.

Generated from `09_builds/recovery_stock_1mb/TECHNICIAN_BRIEF.md` · RP500 project · 2026-11-12 · this document covers only the U26 reflash.